AI 101 for Mechanical Engineers · Module 3

Measurements need context

Lesson 3.2

What you will learn

> Explain why measurements need units and acquisition context, distinguish numerical quantities from categorical identifiers, and assess the limitations of reference data.

Lesson overview

Module 3 · Lesson 2

Measurements need context

AI 101 for Mechanical Engineers

1 Keep metadata with measurements

Metadata explain how data were collected and what they mean.

- Units
- Sensor ID
- Calibration
- Timestamp
- Sampling frequency
- Window length
- Operating state

The same number in W and in kW means a different amount of power.

A raw reading is not a windowed feature

Samples over one second → Vibration RMS feature

RMS summarizes a window of readings.

2 Numbers can be category labels

Machine ID 12 is not twice Machine ID 6

ID numbers identify machines. They do not measure a physical amount.

Some categories have an order (ordinal)

Healthy → Watch → Replace

The order matters, but the steps are not necessarily equal.

Others have no natural order (nominal)

Examples: material type, machine identity.

Assigning numbers to categories does not make them measurements.

3 Ground truth is a reference

A reference value or label can still contain errors.

What can affect it?

- Sensor calibration problems
- Uncertain inspection labels
- Misaligned timestamps

Record provenance

Where did the reference come from, and how was it obtained?

A calibrated measurement can still have uncertainty.

Key takeaway

Keep units, context and reference uncertainty visible when interpreting data.

Metadata improve interpretation but do not establish causation.

AI 101 for Mechanical Engineers · Lesson 3.2

© 2026 Prof. Mazharul Islam · CC BY 4.0

Metadata gives measurements meaning

- Metadata: how data were obtained and what they mean [22]
- Record units, sensor identifier, calibration and timestamp
- Also sampling frequency, window length and operating state
- Same number in W and kW: different magnitudes

A raw sample is not a window feature

Raw time sample

- A single raw reading
- Sampling frequency: samples per unit time

Window feature

- Computed over a time window
- e.g. vibration RMS over one second

> They are different data objects; the feature summarizes a window.

A number can be an identity, not a magnitude

Numerical

- Continuous quantity, e.g. temperature
- Count, e.g. mechanical cycles

Categorical

- Identifies groups, e.g. material grade, machine
- Machine ID 12 is not necessarily twice ID 6

Differences and ratios of identifiers have no physical meaning unless the field represents a quantity.

Some categories have an order

Scale	Example	Order
Ordinal	Healthy, watch, replace	Order matters; gaps not necessarily equal
Nominal	Material type (usually)	No natural order

> Some categories are ordered, others not [23]; numbering does not make them quantities.

Ground truth is a reference, not a guarantee

- Reference measurement or annotation used for comparison
- Miscalibration, uncertain inspection or misaligned timestamps can corrupt it
- A calibrated power measurement can still have uncertainty [24]
- Record provenance: its origin and how it was obtained

> Metadata improves interpretation but does not establish causation.

CHECK YOUR UNDERSTANDING

Why is machine ID 12 not necessarily twice machine ID 6?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Why is machine ID 12 not necessarily twice machine ID 6?

> The numbers identify categories or individual machines. They do not represent a physical quantity for which that ratio has meaning.

CHECK YOUR UNDERSTANDING

Why can a ground-truth target be imperfect?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Why can a ground-truth target be imperfect?

> It can contain measurement or annotation error, including calibration problems, uncertain inspection or timestamp misalignment.

Key takeaways

- Keep units and acquisition metadata with measurements
- Raw readings differ from features computed over time windows
- Categorical identifiers do not imply numerical magnitude
- Check the provenance and uncertainty of reference targets

References

[22] M. D. Wilkinson et al., “The FAIR Guiding Principles for scientific data management and stewardship,” Sci. Data, vol. 3, Art. no. 160018, 2016. doi: 10.1038/sdata.2016.18

[23] S. S. Stevens, “On the theory of scales of measurement,” Science, vol. 103, no. 2684, pp. 677–680, 1946. doi: 10.1126/science.103.2684.677

[24] Joint Committee for Guides in Metrology, “Evaluation of measurement data—Guide to the expression of uncertainty in measurement,” JCGM 100:2008, BIPM, Sèvres, France, 2008. doi: 10.59161/JCGM100-2008E

Numbers match the course reference list. The written lesson is the authoritative text.